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CL-2 (BI) Assignment 6: Agricultural Genomics and Crop Improvement

Analyze crop genomic data to identify genetic markers associated with desirable traits such as disease resistance or yield.

```
import numpy as np, pandas as pd, matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error,mean_squared_error,r2_score
```

```
rng=np.random.default_rng(42)
n_plants,n_markers=500,100
X=rng.integers(0,3,size=(n_plants,n_markers))
markers=[f"Marker_{i+1:03d}" for i in range(n_markers)]
effects=np.zeros(n_markers); effects[[5,18,37,64,82]]=[1.8,-1.4,1.2,1.6,-1.0]
yield_value=50+X@effects+rng.normal(0,2.5,n_plants)
df=pd.DataFrame(X,columns=markers); df["Yield"]=yield_value
display(df.head()); print("Shape:",df.shape)
```

	Marker_001	Marker_002	Marker_003	Marker_004	Marker_005	Marker_006	Marker_007	Marker_008	Marker_009	Marke
0	0	2	1	1	1	2	0	2	0	
1	2	0	2	0	2	2	2	1	1	
2	1	2	1	2	1	0	2	2	0	
3	1	2	0	2	1	0	0	1	0	
4	2	2	0	2	1	1	1	0	2	

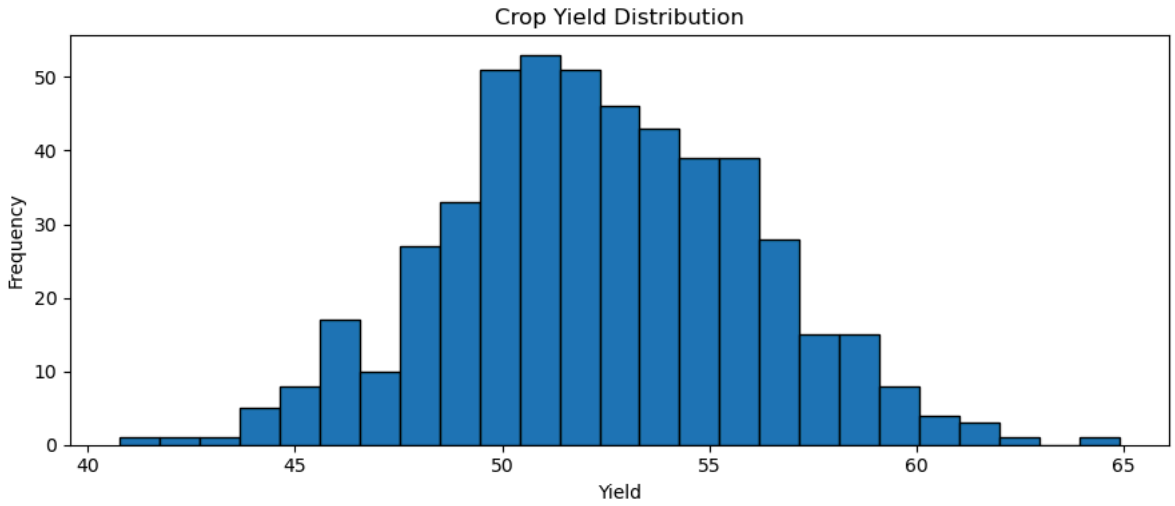
5 rows × 101 columns

```
Shape: (500, 101)
```

```
print("Missing values:",df.isna().sum().sum())
display(df.Yield.describe().to_frame("Yield"))
plt.figure(figsize=(9,4)); plt.hist(df.Yield,bins=25,edgecolor="black")
plt.title("Crop Yield Distribution"); plt.xlabel("Yield"); plt.ylabel("Frequency"); plt.tight_layout(); plt.show(
```

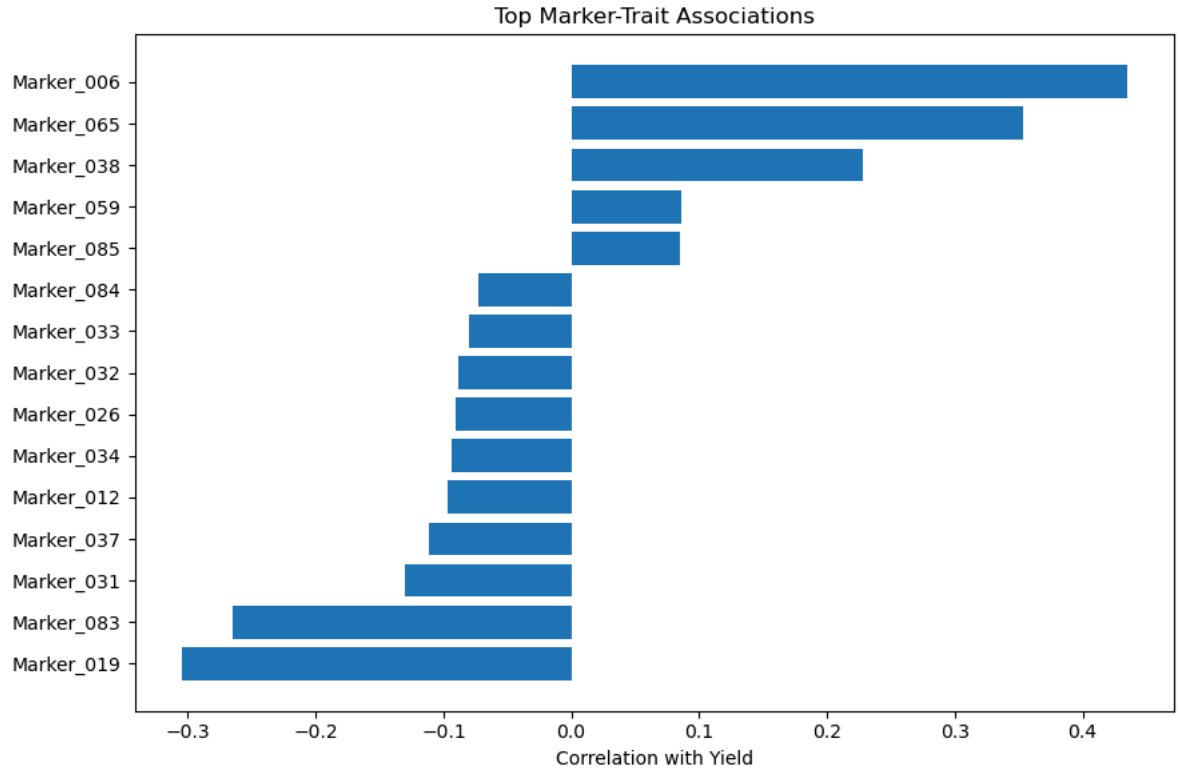
```
Missing values: 0
```

	Yield
count	500.000000
mean	52.354967
std	3.701099
min	40.779222
25%	49.901240
50%	52.201145
75%	54.970736
max	64.890506




```
# Marker-Trait Association Screening
corr=df[markers].corrwith(df.Yield)
assoc=pd.DataFrame({"Marker":corr.index,"Correlation":corr.values,"Absolute_Correlation":np.abs(corr.values)}).s
display(assoc.head(15))
top=assoc.head(15).sort_values("Correlation")
plt.figure(figsize=(9,6)); plt.barh(top.Marker,top.Correlation)
plt.title("Top Marker-Trait Associations"); plt.xlabel("Correlation with Yield"); plt.tight_layout(); plt.show()
```

	Marker	Correlation	Absolute_Correlation
5	Marker_006	0.434544	0.434544
64	Marker_065	0.353566	0.353566
18	Marker_019	-0.304629	0.304629
82	Marker_083	-0.264839	0.264839
37	Marker_038	0.227802	0.227802
30	Marker_031	-0.130347	0.130347
36	Marker_037	-0.111096	0.111096
11	Marker_012	-0.096864	0.096864
33	Marker_034	-0.093227	0.093227
25	Marker_026	-0.090650	0.090650
31	Marker_032	-0.088362	0.088362
58	Marker_059	0.085643	0.085643
84	Marker_085	0.084978	0.084978
32	Marker_033	-0.080471	0.080471
83	Marker_084	-0.072474	0.072474




```
# Random Forest Model Traning and Prediction.
X_train,X_test,y_train,y_test=train_test_split(df[markers],df.Yield,test_size=.2,random_state=42)
model=RandomForestRegressor(n_estimators=300,max_features="sqrt",random_state=42)
model.fit(X_train,y_train); pred=model.predict(X_test)
mae=mean_absolute_error(y_test,pred); rmse=np.sqrt(mean_squared_error(y_test,pred)); r2=r2_score(y_test,pred)
print(f"MAE: {mae:.4f}\nRMSE: {rmse:.4f}\nR²: {r2:.4f}")
```

```
MAE: 2.5393
RMSE: 3.1528
R²: 0.2542
```

```
importance=pd.Series(model.feature_importances_,index=markers).sort_values(ascending=False)
display(importance.head(15).to_frame("Importance"))
importance.head(15).sort_values().plot(kind="barh",figsize=(9,6))
plt.title("Top Markers for Yield Prediction"); plt.xlabel("Random Forest Importance"); plt.tight_layout(); plt.s
```

	Importance
Marker_006	0.060364
Marker_065	0.044303
Marker_019	0.036597
Marker_083	0.030617
Marker_038	0.023554
Marker_084	0.013173
Marker_031	0.012526
Marker_012	0.012205
Marker_034	0.011257
Marker_095	0.010770
Marker_037	0.010193
Marker_077	0.009765
Marker_053	0.009641
Marker_026	0.009584
Marker_072	0.009564

